

A284 – Measurement, Projection and the Origin of Probabilism

Johann Pascher

August 2026

Contents

.1	What this is about	1
.2	Light rays as observer abstraction — the EM field as carrier	1
	What a light ray actually is	1
	Why it is so hard to let go of the light-path picture	2
	Why do lenses work then?	2
	The connection to the Bell question	3
.3	What measurement is in FFGFT	3
	Measurement is a Type-II projection	3
	The geometric reading of collapse	3
.4	Where probabilism comes from	4
	Three possible explanations	4
	Why the FFGFT reading does not violate Bell's theorem	4
	Born rule as projection statistics	4
.5	Three levels of non-locality	4
	Level 1 — Torus picture: topological neighbourhood	5
	Level 2 — Field picture: global eigenmode	5
	Level 3 — Measurement process: global projection	5
	Why evaluation remains probabilistic in principle	5
.6	Field travel time, not path travel time — the measurable delay	6
	The crucial distinction	6
	What propagates in a Bell measurement	6
	The ξ delay as a field phase offset	6
	Why it has never been measured	7
.7	Open question: are A and B the same field point?	9
	The question	9
	Three structurally distinct possibilities	10
	What the corpus already implies	10
	Consequence for the 445-ns prediction	11
.8	The most plausible explanation — a judgement	11
	Group 2 (Doc. 074): Artefact, not alternative	11
	Group 3 (Doc. 023a): Experimentally refuted	11
	Group 4 (Doc. 055): Compatible, but second order	11
	Group 5 (Doc. 142, Jiang PRL 2024): Structurally strongest argument	11
	The overall judgement	12
.9	Open questions	12
.10	Layer-status summary	12
.11	Source references	13
.12	Use of AI tools	14

Abstract. This document closes a gap that A070, A090 and A160 each only hint at: the closed argument for *why* measurement in FFGFT affects the whole system, and *why* evaluation is probabilistic

despite deterministic field equations. The three levels are carried separately and brought together at the end in the context of the instantaneity question of Bell correlations.

Core theses: **(1)** Measurement is a geometric Type-II projection (A090) — it acts on the global state on T^4 , not on a local subsystem. **(2)** Probabilism does not arise from fundamental randomness but from the information loss of the projection: the Born rule is the statistics of Type-II loss. **(3)** What appears as instantaneous action between two measurement points A and B is the statistical signature of a global mode, read out at two projection sites of the same system — no process runs from A to B.

Layer markers. **[STIPULATION]** declared starting point — **[B]** algebraically/mathematically derived — **[K]** numerically computed from corpus constants — **[S]** plausibility sketch — **[H]** open research question.

.1 What this is about

[STIPULATION] In standard quantum mechanics the measurement process is the most conceptually unsatisfying element: a unified, deterministic evolution of the wave function breaks down at measurement, a probabilistic outcome appears, and two distant particles seem to know about each other instantaneously. Over four decades after Bell violation, the question of what physically happens remains open.

[B] FFGFT already has the elements of an answer distributed across other documents: A090 (projection chain, Type I/II/III), A070 (measurement as pole transition, Born rule from phase ignorance), A160 (Bell correlations as topological connection), A030 (winding classes without location). What is missing is the closed chain of argument. This document draws it.

.2 Light rays as observer abstraction — the EM field as carrier

What a light ray actually is

[S] A light ray is not a physical quantity. It is a *geometric abstraction* distilled from the EM field under specific conditions — namely when the wavelength λ is small compared to all relevant geometric dimensions (apertures, objects, observer distance). This condition is *ray optics as a limiting case of wave optics* — mathematically: $\lambda \rightarrow 0$ in the wave equation yields the eikonal equation, from which rays follow as normal trajectories of phase surfaces.

[B] The EM field is the carrier. It exists and evolves according to Maxwell's equations, entirely independent of observers. The light ray appears only when an observer or detector arrangement makes a position measurement: the detector responds to the field at its location and *constructs* backwards a ray as the geometric line connecting source and detector. The ray is the result of this construction — not its precursor.

[B] This is not new — it is fully present in classical wave optics. What FFGFT adds: field evolution on T^4 has no locations in the standard sense. A photon has no trajectory; it has a mode structure on the torus. What we observe as a "light path" is the statistical accumulation of detection events in a spatial region — a projection statistic, not a physical trajectory (A090, Type-II projection).

Why it is so hard to let go of the light-path picture

[S] The light-path picture sits deep for four reasons:

(1) Evolutionary. The human brain is optimised for causality through spatial transfer: stone flies, hits, acts. "Cause before effect, carrier connects, distance separates" — these heuristics were survival-relevant for 10^5 years. They sit deeper than any theory.

(2) Linguistic. Every natural language has subjects performing predicates: “Light goes from A to B.” The idea that a field exists and an observer reacts locally — without “something” travelling from A to B — is not grammatically natural. Even physically correct sentences sound like negations, not positive statements.

(3) Historical. Newton thought in corpuscles — light particles on trajectories. Maxwell solved this: fields instead of trajectories. But then photons arrived, and everyday physics regressed to the trajectory picture: “The photon goes through the double slit.” Wave optics became a calculation method, the particle trajectory became the intuition. This regression is deeply embedded in physics education.

(4) Visibility only via detours. Light rays become visible only through intermediaries: dust, smoke, droplets — scattering centres that locally absorb and re-emit the field. What we see in a searchlight beam is millions of scattering events, each at a different location. The “beam” is the statistical line of these events. Without scatterers the EM field is invisible — physically real, but not an object of observation.

Why do lenses work then?

[B] The lens question is the most precise test of the thesis. If light rays are abstractions and the EM field is the carrier — why does a lens focus?

The answer lies in two layers:

Layer 1 — Wave optics (complete description): The EM field encounters the glass medium of the lens. Inside the glass the phase velocity is $v = c/n$, less than in vacuum. The wave front — a surface of equal phase — slows down more in the thicker part of the lens than in the thinner part. After passing through, the wave front is no longer flat but curved: it converges. The field behind develops toward a focal-point structure — a region of maximum field intensity. No ray is needed: Maxwell’s equations in the inhomogeneous medium describe everything completely.

Layer 2 — Ray optics (limiting-case description): In the limit $\lambda \ll f$ (wavelength $\ll$ focal length), Snell’s law holds: $n_1 \sin \theta_1 = n_2 \sin \theta_2$. Rays are now the normal trajectories of wave fronts — they describe *where the field concentrates*, not where a particle comes from. The lens does not deflect bodies; it reshapes phase surfaces.

[B] The decisive point: the lens acts on the field, not on rays. The ray is a useful geometric description of the result — but the field is the cause. An observer behind the lens sees the image because the field has high intensity at the detector location. The observer *constructs* from this a ray path.

[S] In the FFGFT reading: the photon is a field mode on T^4 . It has no trajectory. The lens modifies the local metric of the EM field — it changes the phase structure of the mode. Behind the lens the probability of detecting the photon at the focal point is maximal. The focal point is not a station on a trajectory but a maximum of the projection probability. That lenses “work” is a statement about the distribution of Type-II projection events in space — not about trajectories.

The connection to the Bell question

[B] The connection to the instantaneity question is direct: whoever asks “how can the measurement at A instantaneously affect B?” is thinking in light paths. The light path is the implicit assumption that every effect has a spatial carrier that moves from A to B.

But if the EM field is the carrier — and the field exists globally on T^4 — then there is no “from A to B”. The field is already everywhere. The measurement at A is a local interaction with the global field. The correlation at B is the response of the same global field at a different projection point. No carrier needs to travel.

[S] The lens example makes this tangible: when light passes through a lens and arrives at the focal point, no one asks “which ray caused the focal point?” That is the wrong question. The field has the phase structure that leads to this focal point. Likewise: when an entangled photon pair is measured, FFGFT does not ask “which signal ran from A to B?” The field has the mode structure that leads to these correlations.

.3 What measurement is in FFGFT

Measurement is a Type-II projection

[B] In the projection chain (A090) there are three qualitatively distinct operations: Type I (duality decompactification, nearly reversible), Type II (geometric projection, lossy, irreversible), Type III (representational translation, bijective).

[B] A measurement is a Type-II operation. It collapses a dimension of the state space — it projects the global state on T^4 onto a lower-dimensional subspace. The kernel of this projection $\ker(\pi) = \mathbb{R}$ contains information no longer present in the image (A090). This lost information is not gone — it is deposited in the measuring apparatus, which thereby becomes part of the total system.

[B] Three consequences follow immediately:

1. **Measurement acts globally.** The T^4 is boundary-free ($\partial T^4 = \emptyset$, A030) and carries global eigenmodes without a distinguished location. A projection at one projection point changes the global mode content — there is no local measurement that leaves the rest of the system untouched.
2. **The apparatus becomes part of the system.** What was previously system state + apparatus state is afterwards a joint state. The boundary is not a physical boundary but a model boundary (interface definition).
3. **Repetition changes the system.** Every subsequent measurement encounters a different global state than the first. The torus “knows” about the first measurement — not through a signal, but because its global mode content has changed.

The geometric reading of collapse

[S] In standard QM the wave function collapses onto an eigenstate at measurement. In the FFGFT reading (A070) this is a deterministic motion in mode space: the (z, r, θ) point is driven to the nearest stable extreme ($z = +1$ or $z = -1$). This is not an instantaneous action at a distance but a local geometric motion at the projection point. The “collapse” is the Type-II projection step itself.

[B] What is lost in the process is the phase θ — the azimuthal information of the mode. It is no longer accessible in the image after measurement, but it is not destroyed: it is encoded in the measuring apparatus (Type-II loss).

.4 Where probabilism comes from

Three possible explanations

There are three structurally distinct answers to the question of why quantum measurements yield probabilistic outcomes:

Standard reading (ontological randomness): $|\psi|^2$ is an ontological probability. The outcome is fundamentally random, not determined by hidden variables (Bell, Kochen-Specker).

Hidden-variable reading (local): The outcome is determined by hidden local parameters. This reading is experimentally excluded by Bell.

FFGFT reading (projection loss): The outcome is determined by the deterministic field equations on T^4 , but the phase θ is not accessible to the observer. Stochasticity arises at the measuring apparatus (the Type-II projection), not in the natural process.

Why the FFGFT reading does not violate Bell's theorem

[B] Bell excludes *local* hidden variables on the projected $\mathbb{R}^3$. The phase θ on T^4 is not a local hidden variable in the sense of the theorem — it is a global feature of the torus mode, not a parameter sitting at a spatial location. Bell's proof assumes a factorisation of the hidden-variable density across the spatial separation. This factorisation does not hold on T^4 because locations on T^4 are defined by their winding classes, not by spatial coordinates (A030).

[B] The quantitative form: the Born rule

$$P_{\text{FFGFT}}(z = +1) = \frac{1+z}{2} = |\alpha|^2 = P_{\text{QM}}(0)$$

follows from the uniform distribution over $\theta \in [0, 2\pi)$ at large statistics (A070). Probabilism is the ensemble average over all phase states — not fundamental randomness, but averaging over what the embedded observer cannot see.

Born rule as projection statistics

[S] The structural thesis: the Born rule $P = |\psi|^2$ is the statistics of Type-II information loss. In a projection $\pi : \mathbb{C}^n \rightarrow \mathbb{C}^{n-1}$ the information of the lost dimension distributes over the ensemble of repeated measurements. The quadratic form arises because the norm in Hilbert space is Euclidean and the projection acts isometrically on the subspace.

[H] The quantitative connection between Type-II loss and the exact form of the Born rule ($|\psi|^2$ rather than $|\psi|$ or $|\psi|^3$) is not worked out. That is the genuine open point — not whether but precisely how the projection statistics enforces the quadratic form.

.5 Three levels of non-locality

[B] The instantaneity question in Bell tests has three levels that must be kept strictly separate. All three are consistently resolvable in the FFGFT framework — in different ways.

Level 1 — Torus picture: topological neighbourhood

[STIPULATION] (A160) Bell correlations on T^4 are a topological phenomenon. What appears in the projected $\mathbb{R}^3$ as action at a distance are points connected by closed paths in the compact picture — topological neighbours.

[B] For terrestrial separations $d \ll L^*$: both measurement points A and B lie on the same torus sheet. The T^4 with the closure scale $L^* = 4\bar{\lambda}_e/\xi_0^{10} \approx 8.70 \times 10^{26} \text{ m}$ is practically flat on terrestrial scales ($d \sim 10^6 \text{ m}$). (*Corrected on 30 Sept. 2026: previously $1.3 \times 10^{26} \text{ m} - 4 \times 3.8616 \times 10^{-13} \text{ m}/(4/30000)^{10} = 8.70 \times 10^{26} \text{ m}$, as in Doc. 312; $1.3\text{--}1.4 \times 10^{26} \text{ m}$ would be $c/H_0 = 2\bar{\lambda}_e/(\pi\xi^{10})$, not L^* .) But: the topological connection runs not via the Euclidean distance*

but via the winding class. An entangled pair shares the same winding class — that is the meaning of the topological connection, independent of the projected separation.

Level 2 — Field picture: global eigenmode

[B] (A030) Winding classes and eigenmodes have no location. An eigenmode is global — it is a property of the entire torus, not of a region. An entangled pair is not a pair of separate objects with a connection but *one* object with two projection sites (A160).

[B] It follows immediately: nothing propagates from A to B. The correlation is identity, not transmission. Signal-freedom is not an additional condition that must be imposed — it follows from the construction.

Level 3 — Measurement process: global projection

[S] This is the deepest level, not explicitly stated in the previous documents.

The measurement at A is not a local disturbance that subsequently propagates to B. It is a Type-II projection of the *global* state on T^4 . This projection changes the global mode content — the joint state of the total system to which both projection sites A and B belong. What appears at B as an “instantaneous effect” is not an effect coming from A, but the changed boundary condition of the global state from which B reads off.

[S] Put differently: measurement at A reads the global mode and modifies it in doing so. The next measurement at B encounters an already modified global mode — not because a signal travelled from A to B, but because both are part of the same global system.

Why evaluation remains probabilistic in principle

[B] All three levels converge on the same point. Evaluation is probabilistic because:

1. Every single measurement is a Type-II projection (loss of the phase θ).
2. The global mode is modified after every measurement — the starting state is never exactly the same a second time.
3. The ensemble over many measurements delivers the phase averaging from which the Born rule follows.

[B] A Bell correlation cannot be read from a single measurement. An ensemble is always required. The ensemble is the statistics of the projection — and the projection is global. Evaluation is therefore probabilistic in principle, not for convenience, but because the single measurement process systematically hides the information.

.6 Field travel time, not path travel time — the measurable delay

The crucial distinction

[B] There are two fundamentally different ways in which a delay $\Delta t = d/c$ can arise:

Path travel time: An object or signal carrier moves from point A to point B. There is *something* that traverses the path. The detector at B registers this something when it arrives. This is the intuitive picture underlying the concept of the light ray.

Field travel time: The field already exists everywhere. At point A something happens — a measurement, a disturbance, an interaction — and this *change in field state* spreads as a wave front. The detector at B does not register an arriving object but a changed field configuration at its location.

[B] The mathematical expression is identical for both cases — $\delta(|\mathbf{x} - \mathbf{x}'| - c(t - t'))$ — but the physical interpretation is fundamentally different. In the Green's function (Doc. 131):

$$G(\mathbf{x}, \mathbf{x}', t - t') = \theta(t - t') \cdot \frac{\delta(|\mathbf{x} - \mathbf{x}'| - c(t - t'))}{4\pi|\mathbf{x} - \mathbf{x}'|}$$

$\mathbf{x}'$ is the *source point of the disturbance* and $\mathbf{x}$ the *observation point*. The δ function says: the change reaches the observation point exactly when $|\mathbf{x} - \mathbf{x}'| = c(t - t')$. What propagates is the *wave front of the field change* — not a carrier.

What propagates in a Bell measurement

[S] At the moment of measurement at detector A the following happens in field terms:

1. The global EM field (the entangled state) interacts with the measuring apparatus at location A. This is a local interaction — not a global event.
2. Through this interaction the field state at A changes: phase, amplitude, mode occupation are modified. This is a *local field change*.
3. This local field change spreads as a retarded wave — with the Green's function $G(\mathbf{x}, \mathbf{x}_A, t - t_A)$. It reaches every point $\mathbf{x}$ at time $t_A + |\mathbf{x} - \mathbf{x}_A|/c$.
4. Detector B at location $\mathbf{x}_B$ registers the changed field configuration at time $t_A + d_{AB}/c$ — not because a signal travelled from A to B, but because the field change has reached it.

[B] This is field travel time, not path travel time. The distinction is physically real:

- With path travel time: B must *wait* until the signal arrives. Before that, B *knows nothing*.
- With field travel time: B interacts with the global field the whole time. Before, B responds to the undisturbed field state. After, B responds to the field state modified by the A measurement. The transition occurs when the wave front arrives.

[B] No information has been transmitted from A to B. The modified field configuration at B is not a message from A — it is the natural evolution of the global field after the A interaction. Since B did not know the unmodified field state anyway (Type-II projection, phase θ unknown), B cannot extract information about the A measurement from the modified field state either — except by comparison with A via a classical channel.

The ξ delay as a field phase offset

[K] (A165) The FFGFT prediction:

$$\Delta t_\xi = \xi \cdot \frac{d}{c} \approx 445 \text{ ns} \quad (d = 1000 \text{ km})$$

[S] This delay is *not* a path travel time and *not* an additional signal transmission. It is a *phase offset in the field wave front* arising from the fractal torus geometry.

Precisely: the global field on T^4 is not completely flat — on terrestrial scales there is a residue of compactness (A090). This residue generates a geometric phase shift of the field modes. When the wave front of the field change spreads out from A, it carries this phase offset — not because it takes a detour, but because the field geometry on the torus deviates slightly from the Euclidean.

[S] This means: what detector B measures is the field amplitude with a phase offset $\xi \cdot d/c$ relative to what B would measure without this geometric effect. The 445 ns is the time over which this phase offset shifts the correlation statistics — not the travel time of a signal.

[B] Whether this effect is measurable depends on whether one can measure the correlation statistics with a time resolution $\ll 445 \text{ ns}$ relative to a common time reference Q.

Measuring the path travel time (signal A→B) would be conceptually wrong. Measuring the field travel time (change in correlation statistics relative to $t_Q + d_B/c$) is the right question.

Why it has never been measured

[K] Preliminary: what has actually been proved and what has not. The four generations of Bell experiments proved one thing: the correlation statistics violates the classical bound ($S > 2$). What they did *not* prove and could not prove: that the correlation arises *instantaneously*. Instantaneity is not an experimental observation but an *interpretation* from the standard QM formalism, in which collapse is by construction timeless. None of the four generations posed the timing question — time resolution was either too coarse (Aspect: μs), or the experiment had no common source-point timestamp (Micius). The statement “Bell correlations are instantaneous” is correct as a *statement of the QM formalism*, but it is *not experimentally verified*.

[K] Comparison with the corpus Bell documents: A160 (topological connection, CHSH, ξ residue) and A165 (hardware validation, 445-ns prediction) treat Bell tests extensively. Doc. 131 (apparent instantaneity) shows that $T \cdot E = 1$ is a *local constraint*, not a dynamic action at a distance, and that the field equations have finite propagation velocity $v \leq c$. None of these documents states explicitly: *the instantaneity of the correlation has never been directly measured, only inferred from the standard QM interpretation*. That is the gap this section closes.

[K] Classification of Doc. 023 (Bell tests in the T_0 context, legacy corpus): Doc. 023 — now superseded by A160 — describes a ξ modification of the Bell correlation:

$$E^{T_0}(a, b) = -\cos(a - b) \cdot (1 - \xi \cdot f(n, l, j))$$

This gives $\text{CHSH}^{T_0} \approx 2.827$ instead of $2\sqrt{2} = 2.828$ — a reduction of $\Delta \approx 10^{-4}$, consistent with A165 ($\Delta\text{CHSH} \approx \xi/(2\pi) \approx 2.1 \times 10^{-5}$; the numerical difference is explained by the different approach via $f(n, l, j)$ factors). The principle is the same in both: FFGFT modifies the correlation by $O(\xi)$. Additionally, Doc. 023 derives an extended formula from ML divergence at $\Delta\theta > \pi$ — a simulation-artefact inference that was never incorporated into the main corpus and remains [S].

Crucially: Doc. 023 treats exclusively the *angular dependence* of the correlation (whether $S > 2$). The timing question — *when* does the correlation arise? — is not posed there either. This confirms the statement of this section: all Bell documents of the corpus, including Doc. 023, have not posed the instantaneity question. A284 adds this dimension — without contradiction to the CHSH corrections.

The complete survey of the legacy documents yields five content groups:

[K] Group 1 — Consistent with main corpus (Docs. 023, A160, A165): CHSH correction $O(\xi)$ below the QM bound: $E^{T_0}(a, b) = -\cos(a - b) \cdot (1 - \xi \cdot f(n, l, j))$, giving $S^{T_0} \approx 2.827 < 2\sqrt{2}$. Prediction: $\Delta\text{CHSH} \approx 10^{-4}$, measurable. Anchored in main corpus (A160, A165) and remains valid.

[S→?] Group 2 — Contradiction with main corpus (Doc. 074): Doc. 074 claims T_0 exceeds the Tsirelson bound $2\sqrt{2}$ by 133 ppm: $|S_{T_0}| = 2.389133$. (Note of 30 Sept. 2026: $2.389133 < 2\sqrt{2} = 2.828427$ does not exceed the Tsirelson bound. Doc. 074 refers the 133 ppm to its value 2.389 for non-optimal angles, which it calls the quantum-mechanical limit (annotated there as well); exceeding $2\sqrt{2}$ by 133 ppm would give $2\sqrt{2}(1+1.33 \times 10^{-4}) = 2.82880$.) It simultaneously uses superdeterminism (violation of the measurement freedom assumption) as the explanatory strategy: apparatus and quantum system are correlated via spatially extended energy fields. This directly contradicts A160: “What does not change: CHSH and the Tsirelson bound.” Two alternatives: (a) The 133-ppm prediction is a computational artefact (Doc. 074 sets $\varepsilon_{T_0, \text{direct}} = \xi$ without gravitational suppression) and is to be discarded.

(b) It is a genuine alternative to A160 — then both are simultaneously falsifiable in the same experiments. Doc. 074 is not incorporated in the main corpus; the superdeterminism framework is not congruent with the topological thesis of A160.

[S→contradiction] Group 3 — Distance-dependent damping (Doc. 023a): $\rho^{T0} = \rho_0 \cdot \exp(-\xi \cdot \Delta d / D_f)$ — entanglement decreases with distance. This is in *direct contradiction* with A160 (winding classes have no location, no distance; entanglement is identity, not transmission). Two structurally distinct frameworks: damping model (Doc. 023a) vs. topological identity model (A160). The main corpus (A160) is authoritative; Doc. 023a is superseded.

[S] Group 4 — Energy-dependent timing difference (Doc. 055): For entangled photons with $\omega_1 \neq \omega_2$:

$$\Delta t_{\text{corr}} = \frac{G\langle m \rangle}{r} \left| \frac{1}{\omega_1} - \frac{1}{\omega_2} \right|$$

Timing statement about *differences within the pair*, not about instantaneity between A and B. Mechanism (Gm/r , gravitational) differs from $\xi \cdot d/c$ (A165, geometric-fractal). Possible bridge: if $G\langle m \rangle/r$ is reducible to ξ via $\xi = 2\sqrt{G} \cdot m$, both predictions would be compatible. This reduction is not worked out in the corpus — open bridge [H].

[Q] Group 5 — External evidence for finite formation time (Doc. 142, Jiang et al., PRL 2024): Jiang, W.-C. et al., *Time Delays as Attosecond Probe of Interelectronic Coherence and Entanglement*, Phys. Rev. Lett. 133, 163201 (2024): quantum entanglement does not arise instantaneously but builds up over ~ 232 attoseconds (theoretical prediction, no laboratory experiment). The only peer-reviewed source in the corpus establishing directly: entanglement formation has a finite time structure. Directly supports A284 — in a different system (helium under EUV laser), but structurally identical.

[K] Overall assessment:

1. Group 1 (Docs. 023/A160/A165): valid, consistent, main corpus.
2. Group 2 (Doc. 074): Tsirelson excess and superdeterminism — contradiction with A160; to be clarified whether artefact or alternative.
3. Group 3 (Doc. 023a): distance-dependent damping — contradiction with A160; main corpus is authoritative.
4. Group 4 (Doc. 055): energy-dependent timing difference — compatible, bridge to ξ still open [H].
5. Group 5 (Doc. 142): external evidence for finite formation time — supports A284.

Previous Bell experiments fall into four generations. None of them posed the question that would be decisive for the FFGFT prediction.

Generation 1: Demonstrating Bell violation. Aspect, Dalibard and Roger (1982) [1] demonstrated Bell violation with time-varying analysers for the first time. The goal was to show $S > 2$. Timing information was used only to identify coincidences (windows $\sim \mu\text{s}$), not to resolve the correlation structure in time.

Generation 2: Closing the locality loophole. Weihs, Jennewein, Simon, Weinfurter and Zeilinger (1998) [2] closed the locality loophole via fast random choice of measurement setting ($\tau_{\text{decision}} \ll d/c$). Time resolution (~ 1 ns) was sufficient for that purpose, but the experiment measured A against B — not relative to the source point Q.

Generation 3: Loophole-free Bell tests. Three independent groups (Hensen et al. 2015 [3]; Giustina et al. 2015 [4]; Shalm et al. 2015 [5]) simultaneously closed locality and detection loopholes. Time resolution: ~ 1 ns, separation $d \sim 1$ km. For $d = 1$ km FFGFT would predict $\Delta t = \xi \cdot d/c \approx 0.44$ ns — just below the time resolution of these experiments. But again the question was: does S violate the classical bound? Not: is there a delay in the correlation structure?

Generation 4: Cosmic separations. Yin et al. (2017) [6] demonstrated Bell violation with the Micius satellite over ~ 1200 km. For this distance FFGFT would predict $\Delta t \approx 530$ ns — well above the detectability threshold of atomic clocks (~ 10 ns). But Micius had *no* precise source-point reference Q. Timestamps were used for coincidence windows, not for an analysis of the causal propagation structure. Moreover Micius did not have three synchronised clocks (Q, A, B) with a precise common time base.

The structural problem. All four generations measure the same thing: whether the correlation statistics over many events violates the classical bound. That is an ensemble question. The FFGFT prediction of the 445 ns delay is instead a question about the *temporal structure* of the field modification per individual event — statistically extracted, but conceptually distinct.

An FFGFT test would require:

1. A precisely time-stamped source point Q (pair generation time to $\ll 1$ ns accuracy).
2. Three synchronised atomic clocks: Q, A, B (common time base to ~ 1 ns over $d \sim 1000$ km).
3. A statistical analysis extracting the delay Δt of the correlation change at B relative to $t_Q + d_B/c$ — not relative to t_A .
4. Sufficient event count to extract $\Delta t \approx 445$ ns from the noise (shot noise $\sim 1/\sqrt{N}$).

No one has performed this experiment because the question — is there a ξ -scaled delay in the propagation of the field modification? — makes no sense in the standard QM framework: in the standard reading nothing propagates; collapse is instantaneous. The question is only meaningful under a theory that assumes a causal field structure behind the correlations.

Experiment	d	Δt_{FFGFT}	Resolution	Question
Aspect 1982	~ 10 m	~ 0 ps	μs	Bell violation
Weihs 1998	400 m	0.18 ns	~ 1 ns	Locality loophole
Hensen 2015	1.3 km	0.58 ns	~ 1 ns	Loophole-free
Micius 2017	1200 km	530 ns	10 ns	Cosmic distance
FFGFT test	1000 km	445 ns	≤ 10 ns	ξ delay?

.7 Open question: are A and B the same field point?

The question

[H] In the projected $\mathbb{R}^3$ two detectors A and B are separated by $d = 1000$ km. In the torus picture:

$$\mathbf{x}_A \equiv \mathbf{x}_B \pmod{L^*}$$

Since $L^* = 4\bar{\lambda}_e/\xi_0^{10} \approx 8.70 \times 10^{26}$ m, the terrestrial separation $d \sim 10^6$ m is negligible: $d/L^* \approx 1.2 \times 10^{-21}$. (*Corrected on 30 Sept. 2026: previously $L^* \approx 1.3 \times 10^{26}$ m and $d/L^* \approx 10^{-20}$ — $10^6/8.70 \times 10^{26} = 1.15 \times 10^{-21}$; the statement is qualitatively unchanged.*) In the torus picture A and B lie in the same fundamental domain — practically at the same location.

[H] The deeper question is not metric but *field-ontological*: are A and B two different points for the entangled field — or the *same* field point, read out at two projection sites?

The corpus has the elements of the question (A030: winding classes have no location; A090: identification $t \sim t + R_{m_i}$; A160: entangled pair is one object with two projection images) but does not state the question explicitly and gives no answer. This is a genuine gap, not a knowledge gap.

Three structurally distinct possibilities

[H] There are three answers that are not equivalent:

Possibility 1 — two points, topologically connected [S]: A and B are different points in the torus picture, but connected by the same winding class. The entangled pair is a mode that encompasses both points. The field modification after the A measurement propagates as a wave front to B. This implies: the ξ delay $\Delta t = \xi \cdot d/c \approx 445$ ns is measurable (§ .2).

Possibility 2 — identical point in the fundamental domain [H]: Since $d \ll L^*$, A and B are practically the same point in the fundamental domain of the torus: $\mathbf{x}_A \approx \mathbf{x}_B$. The entangled mode is fully local within the fundamental domain — there is no separation between them in the torus picture. The correlation is instantaneous in the torus picture; the 445 ns is an artefact of projection.

Possibility 3 — identical point in mode space [H]: The relevant “location variable” is not $\mathbf{x} \in \mathbb{R}^3$ but the winding class $(n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$. Two entangled photons share the same winding class — they are *identical* in mode space. Spatial separation is entirely a projection artefact. Field travel time is then also a concept of the projected space, not of the compact field.

Possibility	Status	Prediction
Two points, topologically connected	[S]	$\Delta t = \xi \cdot d/c$ measurable
Identical in fundamental domain	[H]	445 ns is projection artefact
Identical in mode space	[H]	Field travel time is projection concept

What the corpus already implies

[B] A030 (winding classes without location) and A090 (identification $t \sim t + R_m$) together provide a structure in which “location” for field modes is not a primary concept. This speaks for Possibility 2 or 3 rather than 1.

[B] A160 says the entangled pair is “one object with two projection images”. If it is *one* object, it has *one* field configuration in the torus picture — not two at different locations. The two projection images only arise through the Type-II projection (A090) onto $\mathbb{R}^3$. This speaks for Possibility 3.

[S] If Possibility 3 is correct, this has a strong consequence: the question “why is the correlation instantaneous?” is then not a question about propagation speed — it is a question about why we take the projection of one object at two sites to be two measurements. The answer is not physics but epistemology: the projection image deceives us about the number of objects.

Consequence for the 445-ns prediction

[H] The prediction $\Delta t = \xi \cdot d/c$ from A165 assumes Possibility 1. It is correct if A and B are different field points in the torus picture, between which a field modification can propagate.

If instead Possibility 2 or 3 holds, the prediction shifts conceptually: the effect then does not exist as field travel time between A and B, but as a phase offset in the projection itself — measurable as a timing asymmetry in the correlation statistics, but with a different origin.

[H] This question cannot currently be decided within the FFGFT corpus. It requires a precise formulation of what “location in the field” means for a mode structure on T^4 — a bridge between the topological structure of A030 and the measurement process of A090 that has not yet been built.

Until clarified, A165 remains valid as a prediction — under the explicit assumption of Possibility 1. The assumption is to be declared, not concealed.

.8 The most plausible explanation — a judgement

[S] This section contains a judgement by the author. It is not a proved statement of the corpus but a reasoned assessment based on the analysis in §.2 through §.7.

Group 2 (Doc. 074): Artefact, not alternative

A160 states explicitly as a [STIPULATION] falsification condition: *“If a CHSH value is measured that exceeds the Tsirelson bound, the theory is refuted at this point.”* The Tsirelson bound $2\sqrt{2}$ is an algebraic proof, not an experimental result. Exceeding it would mean refuting standard quantum mechanics itself.

Doc. 074 achieves the 133-ppm excess by dropping the gravitational suppression factor ($G(m)/r \approx 10^{-30}$) and inserting ξ directly. This is an unjustified parameter choice, not a physical argument — Doc. 074 itself calls it an “Alternative Interpretation”.

[S] Judgement: Computational artefact. Discarded.

Group 3 (Doc. 023a): Experimentally refuted

A160 establishes **[B]** algebraically: the $\mathbb{C}^3$ tensor factor is invisible to correlation quantities — the observable acts as identity there (A070). Distance-dependent damping would have measurable consequences: entanglement would decrease with separation. Micius 2017 shows Bell violation over 1200 km at the same level as laboratory experiments. This is experimentally excluded.

[S] Judgement: Experimentally refuted. Discarded.

Group 4 (Doc. 055): Compatible, but second order

In natural units $\xi = 2\sqrt{G} \cdot m$, hence $G = \xi^2/(4m^2)$. The gravitational term of Doc. 055 is therefore $\propto \xi^2$ — second order. The predicted time difference is $\sim 10^{-30}$ s, experimentally inaccessible.

[S] Judgement: Consistent with the ξ scheme at second order, many orders of magnitude below A165. Compatible, but in the noise.

Group 5 (Doc. 142, Jiang PRL 2024): Structurally strongest argument

The Jiang work is independent of the FFGFT framework. It shows from the time-dependent Schrödinger equation for helium that entanglement has a finite build-up time. This is not an FFGFT-specific result but a general finding of time-dependent quantum mechanics.

[S] Judgement: Quantum mechanics itself excludes strict instantaneity when the time-dependent evolution is taken seriously. Standard QM introduces instantaneity through the stationary eigenstate basis, not through the time-dependent dynamics. This is structurally identical to the field travel-time thesis of A284 — in a different system, but with the same mechanism.

The overall judgement

[S] The most plausible explanation for Bell correlations in FFGFT is **Group 1 and Group 5 combined**:

- The correlation does *not* arise instantaneously (Jiang, [Q]) — it has a finite build-up time in the field geometry.
- In the compact torus picture this is no action at a distance, because A and B are projection sites of the same global field (A160, [B]).
- The CHSH correction lies at $O(\xi) \approx 10^{-4}$ below the Tsirelson bound (A165, [K]).
- Distance-dependent damping (Group 3) is experimentally excluded; Tsirelson excess (Group 2) is a computational artefact.

[H] What remains open: whether A and B are the same field point in the torus picture (§ .7). This question decides whether the 445-ns delay (A165) has a measurable consequence — or is itself a projection artefact, because there is no field separation between A and B.

[S] This is the honest remainder. The qualitative answer is clear: no action at a distance, no instantaneity, no distance dependence. The quantitative prediction (445 ns) depends on the unresolved question of whether field points on T^4 carry a well-defined separation at all.

.9 Open questions

[H] Three genuine open points:

1. **Born rule from Type-II loss:** The quantitative derivation of the quadratic form $P = |\psi|^2$ from projection loss is not worked out. Why square and not absolute value or higher power? The isometric property of the Hilbert-space projection is the candidate (A090), but the proof is missing.
2. **Path assignment for entanglement:** Which closed paths on T^4 correspond to which concrete entangled states (A160, [S]). Without this assignment the topological thesis remains a structural statement, not an operational criterion.
3. **Global mode and local measurement:** How precisely does a Type-II projection at one projection point modify the global mode content? The qualitative mechanism is clear (Type-II loss); the quantitative field equation for the global back-action is not worked out.

.10 Layer-status summary

Statement	Status
Measurement is Type-II projection (A090)	[B]
$\partial T^4 = \emptyset$: boundary-free, global eigenmodes without location	[B]
Measuring apparatus becomes part of the total system	[B]
Winding classes have no location (A030)	[B]
Born rule: $P = (1 + z)/2 = \alpha ^2$ from phase averaging	[B]
Bell: no local hidden variables on $\mathbb{R}^3$	[B]
FFGFT: θ is global feature, not local parameter	[B]
Entangled pair: one object, two projection sites (A160)	[B]
Signal-freedom: consequence of construction, not extra condition	[B]
Spatial delay $\Delta t = \xi \cdot d/c$ (A165)	[K]
Measurement at A modifies global mode content	[S]
Born rule as statistics of Type-II loss	[S]
ξ delay as field modification, not signal	[S]
Born rule quadratic form from projection	[H]
Path assignment entangled state $\leftrightarrow T^4$ path	[H]

Statement	Status
Quantitative field equation for global back-action	[H]

Balance: $9 \times [\mathbf{B}] \cdot 1 \times [\mathbf{K}] \cdot 3 \times [\mathbf{S}] \cdot 3 \times [\mathbf{H}]$.

.11 Source references

- A030** Pascher, J., *Compact Geometry*. A-series, 2026. Winding classes without location, $\partial T^4 = \emptyset$.
- A070** Pascher, J., *Hilbert-Space Translation*. A-series, 2026. Measurement as pole transition, Born rule from phase ignorance, bijectivity.
- A090** Pascher, J., *Projection Chain*. A-series, 2026. Type-I/II/III operations, reversibility, information loss.
- A160** Pascher, J., *Quantum Mechanics and Bell Correlations*. A-series, 2026. Topological connection, CHSH, ξ residue.
- A165** Pascher, J., *Bell Hardware and Quantum Computing*. A-series, 2026. $\Delta t = \xi \cdot d/c$, IBM Heron measurements.
- Doc. 131** Pascher, J., *Apparent Instantaneity*. Legacy corpus, 2025. Green's function, causality, time scales.

Bibliography

- [1] Aspect, A., Dalibard, J., Roger, G. (1982). *Experimental test of Bell's inequalities using time-varying analyzers*. Physical Review Letters, 49(25), 1804–1807.
- [2] Weihs, G., Jennewein, T., Simon, C., Weinfurter, H., Zeilinger, A. (1998). *Violation of Bell's inequality under strict Einstein locality conditions*. Physical Review Letters, 81(23), 5039–5043.
- [3] Hensen, B. et al. (2015). *Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres*. Nature, 526, 682–686.
- [4] Giustina, M. et al. (2015). *Significant-loophole-free test of Bell's theorem with entangled photons*. Physical Review Letters, 115(25), 250401.
- [5] Shalm, L. K. et al. (2015). *Strong loophole-free test of local realism*. Physical Review Letters, 115(25), 250402.
- [6] Yin, J. et al. (2017). *Satellite-based entanglement distribution over 1200 kilometers*. Science, 356(6343), 1140–1144.
- [7] Bell, J. S. (1964). *On the Einstein Podolsky Rosen paradox*. Physics Physique Fizika, 1(3), 195–200.

.12 Use of AI tools

This document was produced with the aid of AI-assisted tools. The questions, the stipulations and all decisions of content are the author's; the tools draft the prose, check the arithmetic and produce the verification scripts. Responsibility for content and for errors rests entirely with the author. The full wording is given in A010.